

Exhaust Gas Temperature 1 Signal for LLOYD's Malfunction

Symptoms

- The OEM signal for LLOYD's sensor has malfunctioned.

How To Use This Tree

This symptom tree can be used to troubleshoot a malfunction. Start by performing Step 1 troubleshooting. Step 2 will ask a series of questions and will provide a list of troubleshooting steps to perform, depending on the symptom.

Shoptalk

This LLOYD's sensor is connected to the OEM side (X7 connector) of the remote input/output unit. The OEM is responsible for this connection.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the OEM wiring harness.	
	STEP 1A. Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for an open.	
	STEP 1B. Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for a wire-to-wire short.	
	STEP 1C. Check the exhaust gas temperature 1 signal wire for a short to ground.	
	STEP 1D. Check the exhaust gas temperature 1 sensor supply +24-VDC for voltage.	

STEP 1. Check the OEM wiring harness.

STEP 1A. Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for an open.

Conditions :

- Disconnect the OEM harness at the X7 connector.
- Disconnect the exhaust gas temperature 1 sensor connector.

Action	Specification/Repair	Next Step
Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for an open. • Place one test lead on the exhaust gas temperature 1 signal pin at the X7 connector. • Place the other test lead on the exhaust gas temperature 1 signal pin at the sensor connector. • Place one test lead on the exhaust gas temperature 1 sensor supply +24-VDC pin at the X7 connector. • Place the other test lead on the exhaust gas temperature 1 sensor supply +24-VDC pin at the sensor connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES	2B
	Less than 10 ohms? NO Repair : Replace the wire. Refer to the OEM service manual.	Repair complete

STEP 2B. Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for a wire-to-wire short.

Conditions :

- Disconnect the OEM harness at the X7 connector.
- Disconnect the exhaust gas temperature 1 sensor connector.

Action	Specification/Repair	Next Step
Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for a wire-to-wire short. • Place one test lead on the exhaust gas temperature 1 signal pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. • Place one test lead on the exhaust gas temperature 1 sensor supply +24-VDC pin at the X7 connector. • Place the other test lead on all other pins at the X7 connector. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO	2C

STEP 2C. Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for a short to ground.**Conditions :**

- Disconnect the OEM harness at the X7 connector.
- Disconnect the exhaust gas temperature 1 sensor connector.

Action	Specification/Repair	Next Step
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Check the exhaust gas temperature 1 signal and sensor supply +24-VDC wires for a short to ground. • Place one test lead on the exhaust gas temperature 1 signal pin at the X7 connector. • Place the other test lead on engine ground. • Place one test lead on the sensor supply +24-VDC pin at the X7 connector. • Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	Less than 10 ohms? YES Repair : Replace the wire. Refer to the OEM service manual.	Repair complete
	Less than 10 ohms? NO	2D

STEP 2D. Check the exhaust gas temperature 1 sensor supply +24-VDC wire for voltage.

Conditions :

- Disconnect the exhaust gas temperature 1 sensor connector.

Action	Specification/Repair	Next Step

Check the exhaust gas temperature 1 sensor supply +24-VDC wire for voltage. - Place one test lead on the exhaust gas temperature 1 sensor supply +24-VDC pin at the sensor connector. - Place the other test lead on engine ground. Refer to the appropriate circuit diagram or wiring diagram for pin and wire identification.	+24-VDC? YES Repair : Replace the exhaust gas temperature 1 sensor. Refer to the OEM service manual or contact a Cummins® Authorized Repair Location.	Repair complete
	+24-VDC? NO Repair : Check the batteries. Refer to the OEM service manual. Replace the remote input/output unit. Contact a Cummins® Authorized Repair Location.	Repair complete